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### Implant for bone fixation

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#### Abstract

An implant for bone fixation is presented. The implant comprises an elongated member configured to be wound around bone parts that are to be fixed. Further, an engagement feature is provided that is coupled to the elongated member and configured to engage a portion of the elongated member so as to secure the implant in a loop around the bone parts. A fastening member is arranged between and coupled to the elongated member and the engagement member. The fastening member comprises at least one opening for receiving a bone fastener. In one variant, at least one visual indicator indicating a region of the fastening member to be severed in order to detach the implant is provided. In another variant, at least one of the engagement member and the elongated member are coupled to the fastening member by injection molding.

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**Inventors:** Knoepfle; Christian (Donaueschingen, DE), Greiner; Karl (Muehlheim, DE), Schmuck; Manfred (Muehlheim, DE)

**Applicant:** Stryker European Operations Holdings LLC (Kalamazoo, MI)

**Family ID:** 1000008780502

**Assignee:** Stryker European Operations Holdings LLC (Portage, MI)

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. application Ser. No. 16/374,942, filed Apr. 4, 2019, which is a continuation of U.S. application Ser. No. 14/991,181, filed Jan. 8, 2016, now U.S. Pat. No. 10,292,742, which claims priority from European Patent Application No. 15 000 033.9, filed Jan. 9, 2015, the disclosures of which are each incorporated herein by reference.

### TECHNICAL FIELD

(1) The present disclosure generally relates to bone fixation. Specifically, the disclosure relates to an implant for fixing bone parts, to systems comprising the implant, and to methods of manufacturing the implant.

### BACKGROUND OF THE INVENTION

(2) Various surgical procedures require the surgeon to access the thoracic region of a patient. A known approach to access the thoracic region is to cut the sternum in two parts and separate these two parts from each other for gaining access to the thoracic region. After completion of the surgical procedure, the separated parts of the sternum are brought back to their initial positions and fixed, for example, with a bone plate attached to the sternum parts or a wire tensioned around the circumference of the sternum.

(3) U.S. Pat. No. 5,417,698 discloses a closure element to be looped around a human sternum. The closure element comprises a strap which is inserted through and retained by a tightening plate.

(4) U.S. Pat. No. 8,460,295 discloses a sternum repair device including a central body and a plurality of bands extending from the central body. The bands are wrapped around the sternum to keep the sternum parts together. The central body includes a view window which is used by a surgeon to line up the device during installation on the sternum.

(5) U.S. Pat. No. 8,486,114 discloses a cerclage system including a cable that encircles the sternum parts and a bone plate having channels to receive segments of the cable. The bone plate further includes a pair of locking studs to lock the cable within the channels to the bone plate.

(6) EP 0 608 592 B1 discloses an assembly for banding a sternum. The assembly comprises an elongated flexible band, a needle at one end of the band and a buckle proximate the other end of the band. A main section of the band includes a plurality of spaced apart slots which can engage at a locking mechanism.

(7) After the thoracic procedure such as, for example, a bypass operation has been carried out on a patient and the sternum parts have been fixed using a suitable fixation assembly, the patient is normally kept under surveillance. If it is detected that the surgical procedure has failed or that complications occur, it may be desirable for the surgeon to again open the fixation system for accessing the thoracic region. The time required for this opening procedure may be critical for the patient's health and even life.

(8) In other situations, a cable or wire tensioned around the sternum parts (so-called "primary closure") might become loose or break due to the load applied to the thoracic region of the patient. In such cases, the sternum part fixation may need to be stabilized by, for example, a bone plate (so-called "secondary closure").

(9) Similar or related problems occur in connection with other surgical procedure in which two or more bone parts need to be fixed.

(10) There is a need for an implant for bone fixation that can be attached or separated fast, easily and at low risk for the patient. Furthermore, there is a need for such an implant which provides a high implant-bone-construct-stability. Moreover, there is generally a need for an implant for bone



fixation that exhibits good surgical results.

## SUMMARY OF THE INVENTION

(11) According to a first aspect of the present invention an implant for bone fixation is provided. The implant comprises an elongated member, an engagement member, a fastening member, and at least one visual indicator. The elongated member is configured to be wound around bone parts that are to be fixed and the engagement member is coupled to the elongated member and configured to engage a portion of the elongated member so as to secure the implant in a loop around the bone parts. The fastening member is arranged between and coupled to the elongated member and the engagement member, the fastening member comprising at least one opening for receiving a bone fastener. The at least one visual indicator indicates a region of the fastening member to be severed in order to detach the implant.

(12) In some implementations, at least the elongated member and the fastening member may be made of different materials. As an example, the elongated member may be made from a material that exhibits a certain flexibility so that the elongated member can be wound around the bone parts. On the other hand, the fastening member may be made from a rigid material that will not, or not easily, deform upon an engagement with the bone fastener. As such, the rigidity of the fastening member material may generally be higher than the rigidity of the engagement member material.

(13) In a first realization, the region of the fastening member indicated by the visual indicator may be a region of reduced mechanical strength of the fastening member. As such, the mechanical strength of the fastening member may be lower in the region indicated by the visual indicator than in other regions of the fastening member. The mechanical strength may be defined in relation to a force required to sever (cut, saw, etc.) the fastening member. In another realization, that may be combined with the first realization, the region of the fastening member indicated by the visual indicator may be a region that is to be severed from a surgical viewpoint (e.g., so as to separate the bone parts).

(14) The fastening member may be coupled to at least one of the elongated member and the engagement member by injection molding. In one variant, the fastening member comprises a first end and a second end opposite to the first end. The elongated member may be injection molded to the first end of the fastening member. Additionally, or as an alternative, the engagement member may be injection molded to the second end of the fastening member. As an example, the elongated member may be injection molded to the first end of the fastening member in a first injection molding process and the engagement member may be injection molded to the second end of the fastening member in a second injection molding process separate from the first injection molding process. The first and second injection molding processes may be performed at separate points in time (e.g., one after the other) or substantially at the same point in time (e.g., simultaneously).

(15) If injection molding is used for manufacturing the implant, the visual indicator may be defined by a portion of the fastening member that remains exposed from injection molding. As an example, the exposed region may be located between the first end of the fastening member associated with the first injection molding process and the second end of the fastening member associated with the second injection molding process. The exposed region may have a width of 1 to 10 mm.

(16) Regardless of the manufacturing process, the visual indicator may also be realized in any other manner. As an example, the visual indicator may be printed on the implant. Additionally, or in the alternative, the visual indicator may be realized as a three-dimensional structure, for example so as to comprise a notch or a groove. As will be appreciated, such a three-dimensional structure could also facilitate severing of the fastening member.

(17) The visual indicator may in one variant extend in a direction substantially perpendicular to an extension of the elongated member. For example, the visual indicator may indicate a direction in which the fastening member is to be severed. In another variant, the visual indicator may have no preferred extension (e.g., it may take the form of a point, circle, or similar non-directed feature).

(18) The fastening member may comprise a plurality of openings. In this regard, the fastening

member may comprise at least a first opening and a second opening that are substantially aligned with an extension of the elongated member. The visual indicator may be located between the first opening and the second opening. In variants in which the fastening member comprises more than two openings, still a single visual indicator may be provided (e.g., separating the plurality of openings in two sets of two or more openings each).

(19) In a fastening member with several openings, the first opening may be configured to be located proximate to a first bone portion and the second opening may be configured to be located proximate to a second bone portion to be fixed. There may be further openings located proximate to the first and second bone portions, respectively (e.g., to insert two or more bone fasteners in each bone portion).

(20) In a second aspect of the present invention, a further implant for bone fixation is provided. The implant comprises an elongated member configured to be wound around bone parts that are to be fixed, an engagement member coupled to the elongated member and configured to engage a portion of the elongated member so as to secure the implant in a loop around the bone parts, and a fastening member arranged between and coupled to the elongated member and the engagement member. The fastening member comprises at least one opening for receiving a bone fastener and is coupled to at least one of the elongated member and the engagement member by injection molding.

(21) The implant of the second aspect may comprise a visual indicator as generally discussed herein. In other variants, no such visual indicator may be provided.

(22) In all the implant aspects presented herein, the elongated member may comprise a plurality of engagement features provided along an extension of the elongated member. In such a case, the engagement member may be configured to be brought into selective engagement with one of the engagement features. In one variant, the engagement features and the engagement member realize a zip tie-type engagement mechanism.

(23) In all the implant aspects, a hook member may be coupled to the elongated member at an end thereof opposite to the engagement member. The hook member may be coupled to the elongated member by injection molding or may form an integral part of the elongated member.

(24) In all the implant aspects, the at least one opening of the fastening member may comprise a locking feature configured to lock the bone fastener to the fastening member. Further, the locking feature may be configured to engage the bone fastener at a selected angular orientation. The locking feature may take the form of a circumferential lip (e.g., of increased material thickness) or of a thread. The at least one opening of the fastening member may comprise a threaded portion on a bone facing side of the fastening member and an unthreaded portion on a side opposite to the bone facing side.

(25) A portion of the fastening member adjacent to the at least one opening may have a conical, convex or spherical taper which substantially tapers inwardly in a direction toward a bone facing surface of the fastening member. Thus, the fastening member can be configured to exert a compression force when a bone fastener is screwed or inserted through the opening into a bone part.

(26) In one implementation, the at least one opening of the fastening member may be adjacent to an inclined surface onto which a bone fastener is able to slide in a fastening or compression position. The inclined surface may permit a bone fastener to slide laterally or longitudinally with respect to the opening or the fastening member. Further, at least one opening of the fastening member may define a predetermined direction for a bone fastener. The inclined surface may have a predetermined angle with respect to an extension plane of the attachment member. The predetermined angle can be between about 20 and 70 degrees, for example about 40 to 50 degrees (e.g., about 45 degrees).

(27) The at least one opening of the fastening member may generally be a circular or elongated hole. The elongated hole may be an oblong hole. Further, the elongated hole may extend substantially parallel with respect to a longitudinal direction of the fastening member or of the

flexible elongated member. Alternatively, the elongated hole may extend substantially perpendicular to a longitudinal direction of the fastening member or of the elongated member.

(28) The implant may have a larger width in a region of the one or more openings compared to a general width of the elongated member. The thickness of the fixation assembly may be larger in a region of the one or more openings compared to a general thickness of the elongated member.

(29) In all the implant aspects, the fastening member may be made from a metallic material (e.g., stainless steel, titanium or a titanium alloy). In addition, or as an alternative, at least one of the elongated member and the engagement member may be made from a polymeric material (e.g., polyetheretherketone, PEEK). In certain variants, the engagement member and the fastening member may at least partially be realized as a one-piece structure. As such, the engagement member may at least partially be made from the same material than the fastening member.

(30) Also provided is a an implant system comprising the implant presented herein and at least one bone fastener configured to be inserted through the at least one opening of the fastening member into bone.

(31) As said, the at least one opening may comprise a first locking feature configured to lock the bone fastener to the fastening member. In such a case the bone fastener may comprise a second locking feature configured to engage the first locking feature. As an example, the first locking feature may take the form of a circumferential lip or thread. The second locking feature may take the form of a thread. This thread may be provided in a head region of the bone fastener or immediately below the head region. The second locking feature may be different from a bone engaging thread of the bone fastener.

(32) Still further, a method of manufacturing an implant is provided, wherein the implant has an elongated member configured to be wound around bone parts that are to be fixed, an engagement member coupled to the elongated member and configured to engage a portion of the elongated member so as to secure the implant in a loop around the bone parts, and a fastening member arranged between and coupled to the elongated member and the engagement member, the fastening member comprising at least one opening for receiving a bone fastener. The method comprises coupling the fastening member to at least one of the elongated member and the engagement member by injection molding.

(33) The method aspect may comprise one or more further steps or substeps to manufacture the implant presented herein.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other features, aspects and advantages of the present disclosure will now be explained with reference to the following description of exemplary embodiments and the accompanying drawings, wherein:
- (2) FIG. 1 is a perspective view of an embodiment of an implant for bone fixation;
- (3) FIG. 2 is an enlarged perspective view of a portion of the implant of FIG. 1;
- (4) FIG. 3 is a perspective cross-sectional view of the implant portion shown in FIG. 2;
- (5) FIG. 4 is a perspective view of a fastening member for use in connection with the implant of FIG. 1;
- (6) FIG. 5 is a perspective cross-sectional view of the fastening member shown in FIG. 4;
- (7) FIG. 6 is a schematic cross-sectional view illustrating an angular orientation of a bone fastener relative to a fastening member in accordance with a further embodiment;
- (8) FIG. 7 is an enlarged perspective view of another portion of the implant of FIG. 1; and
- (9) FIG. 8 is a perspective view of a hook member of the implant of FIG. 1.

### DETAILED DESCRIPTION

(10) In the following description, exemplary embodiments of a bone fixation implant and an implant system comprising the implant and a bone fastener will be explained with reference to the drawings. The same reference numerals will be used to denote the same or similar structural features.

(11) FIG. 1 shows a perspective view of an embodiment of a bone fixation implant **10**. In the embodiment shown in FIG. 1, the implant **10** is configured and dimensioned for fixing bone parts of a sternum. It will be appreciated that the present disclosure is not limited to this surgical indication. Rather, the present disclosure can also be applied in connection with fixing bone parts in other regions of the human anatomy, possibly with suitably adapted configurations and dimensions.

(12) The implant **10** shown in FIG. 1 comprises an elongated member **12**, an engagement member **14** coupled to the elongated member **12**, and a fastening member **16** arranged between and coupled to the elongated member **12** and the engagement member **14**.

(13) The elongated member **12** is configured to be wound around the sternum parts that are to be fixed. As such, the elongated member **12** is made from a generally flexible material and has a suitable length of approximately 10 to 35 cm. The engagement member **14** is configured to engage a portion of the elongated member **12** so as to secure the implant **10** in a loop around the sternum parts. Details of how the engagement member **14** engages the elongated member **12** will be described in more detail below.

(14) As also shown in FIG. 1, the implant **10** further comprises a hook member **18** coupled to the elongated member **12** at an end thereof opposite to the engagement member **14**. The hook member **18** is configured to facilitate threading of the elongated member **12** around the sternum parts that are to be fixed. It will be appreciated that depending on the surgical indication, the hook member **18** could also be omitted or substituted by another member (e.g., a linear needle) to support a proper placement of the elongated member **12** by a surgeon.

(15) In the following, the configuration of the implant portion comprising the engagement member **14** and the fastening member **16** will be described in more detail with reference to FIGS. 2 and 3. As illustrated therein, the fastening member **16** comprises a plurality of openings **20**. Each opening **20** is configured to receive a bone fastener such as a bone screw or a bone pin. The openings **20** are substantially aligned with an extension of the elongated member **12**. In the exemplary case of openings **20** having a circular shape, the centers of the openings **20** lie on an axis that coincides with an axis of the elongated member **12** in a non-wound (i.e., planar) state.

(16) The fastening member **16** of the present embodiment comprises four openings **20**. It will be appreciated that more or less openings **20** could be provided in other embodiments depending on the surgical needs.

(17) As illustrated in FIGS. 2 and 3, the four openings **20** are divided by a visual indicator **22** into two sets of two openings **20** each. The visual indicator **22** indicates a region of the fastening member **16** to be severed in order to detach the implant **10** after implantation. In the present embodiment, the visual indicator **22** has a generally rectangular shape with a larger extension that extends in a direction substantially perpendicular to the extension of the elongated member **12**. The visual indicator **22** thus indicates the direction in which the fastening member **16** is to be severed.

(18) The visual indicator **22** in the present embodiment marks a region of reduced mechanical strength of the fastening member **16** (explained in more detail below with reference to FIG. 4). As such, the visual indicator **22** indicates the region of the fastening member **16** that is easiest to sever by a surgical cutter in order to detach the implant **10**.

(19) In the context of fixing two sternum parts, the visual indicator **22** of the implant **10** after implantation will generally lie on a line along which the two sternum parts have been joined. That joining line will generally extend perpendicular to the extension of the fastening member **16**. For this reason the visual indicator **22** separates a first set of two openings **20** that will be located proximate to one sternum portion from another set of two openings **16** that will be located proximate to the other sternum portion. Severing the fastening member in the region of the visual

indicator **22** will therefore permit to separate the two sternum parts (e.g., in the case of medical complications) even when the bone fasteners attached via the openings **20** to the sternum parts remain in place.

(20) It will be evident that severing the fastening member **16** in the region of the visual indicator **22** in order to detach the implant **10** provides a much quicker access to the thoracic region than detaching the implant **10** by removing multiple bone fasteners inserted to the openings **20** into the sternum parts. It should be noted that after severing the fastening member **16** it may still be required to also sever the elongated member **12** or another portion of the implant **10** to fully separate the sternum parts. In any case, the visual indicator **22** guides the surgeon towards the optimal fastening member region to be severed in order to quickly detach the implant **10**.

(21) The visual indicator **22** may generally be provided in many different ways, for example by printing it on the implant **10** (e.g., directly on the fastening member **10**), by stamping a line or any other three-dimensional pattern in the implant **10** (e.g., directly in the fastening member **16**), or in other ways. In the present embodiment, the visual indicator **22** is realized during the manufacturing process of the implant **10** by exposing a portion of the fastening member **16** during an injection molding process.

(22) Specifically, as illustrated in FIG. 3, the fastening member **16** is coupled to the elongated member **12** and the engagement member **14** by injection molding. In this regard, the elongated member **12** is injection molded to a first end of the fastening member **16**, while the engagement member **14** is injection molded to an opposite end of the fastening member **16**. The visual indicator **22** is thus defined by a central portion of the fastening member **16** that remains exposed from injection molding.

(23) In certain variants, the elongated member **12** is injection molded to the fastening member **16** in a first injection molding process and the engagement member **14** is injection molded to the fastening member **16** in a second injection molding process separate from the first injection molding process. The two injection molding processes may be performed essentially at the same point in time or at spaced apart points in time. It will further be appreciated that in certain embodiments the visual indicator **22** may be omitted. In such a case, or in other cases, the fastening member **16** may be injection molded to the elongated member **12** and to the engagement member **14** in a single injection molding process.

(24) As will be appreciated, the fastening function of the fastening member **16** can best be fulfilled in case the fastening member **26** is made from a rigid material, such as metal (e.g., stainless steel, titanium, or a titanium alloy), especially if compression forces are to be applied via the bone fasteners inserted through the openings **20**. On the other hand, the elongated member **12** will be manufactured from an at least partially flexible material since it needs to be wound around the bone parts that are to be fixed. As such, the elongated member can be made from a polymeric material such as PEEK.

(25) The engagement member **14** may be made from the same material as the elongated member **12** or from a different material. As an example, the engagement member **14** may also be made from PEEK. In another variant, the engagement member **14** may at least partially be made from the same material as the fastening member **16**. As an example, the engagement member **14** and the fastening member **16** may at least partially be realized as a one-piece structure.

(26) As will be appreciated, using a polymeric material for the elongated member **12** and the engagement member **14** facilitates injection molding of those two members **12**, **14** to the fastening member **16**. However, the present disclosure in relation to the visual indicator **22** is not limited to manufacturing the implant **10** using one or more injection molding processes. Moreover, the visual indicator **22** could also be realized in connection with an implant **10** that is realized as a one-piece structure (e.g., from a polymeric material and, optionally, in a single injection molding process).

(27) FIGS. 2 and 3 also illustrate the engagement mechanism that allows the engagement member **14** to engage a portion of the elongated member **12** so as to secure the implant **10** in a loop around

the sternum parts. In more detail, the elongated member **12** comprises a plurality of engagement features **24** that are provided along essentially the entire length of the elongated member **12** (see FIG. 1). As can best be seen in FIG. 3, each individual engagement feature **24** has a sawtooth-like cross-section with a first surface that extends generally perpendicular to a plane defined by the elongated member **12** in an un-wound state, and a second surface at an angle to that plane.

(28) The engagement member **14** comprises complementary engagement features **26** with a sawtooth-like profile. As shown in FIG. 3, those complementary engagement features **26** are provided on a deflectable pawl **28**. The pawl **28** is located within an opening **30** that extends through a head portion **32** of the engagement member **14**. The opening **30** has an axis that is substantially parallel to a center axis of each of the openings **20** in the fastening member **16**.

(29) The deflectable pawl **28** is attached to the head portion **14** at a bone facing side of the head portion **32** and configured to engage the engagement features **24** of the elongated member **12** in a zip tie-like manner upon threading the elongated member **12** through the opening **30** in the head portion **32** of the engagement member **14**. The zip tie-like configuration of the engagement mechanism permits to generate a compression force when the implant **10** is looped around the sternum parts. This compression force presses the sternum parts against each other and facilitates bone healing.

(30) It will be appreciated that in other embodiments other engagement mechanisms could be used. As an example, the elongated member **12** and the engagement member **14** could each also take the form of a band or wire that can be brought into engagement via knot or a dedicated coupling feature.

(31) In the following, the configuration of the fastening member **16** will be described in more detail with reference to FIGS. 4 and 5. As shown therein, the fastening member **16** generally takes the form of an elongated bone plate with an undulating outer profile. Specifically, the undulations in the outer profile correspond to the location of the openings **20** in the fastening member **16**. As can also be seen in FIG. 4, the central portion of the fastening member **16** (i.e., the region between a first set of two adjacent openings and a second set of two adjacent openings) has a reduced mechanical strength to facilitate severing of the fastening member **16**. In more detail, the central portion of the fastening member **16** is the region of the fastening member **16** with the smallest width. As illustrated in FIGS. 2 and 3, that fastening member region with the smallest width corresponds to the location of the visual indicator **22**.

(32) As also shown in FIGS. 4 and 5, the fastening member **16** comprises a conically tapering portion **34** towards each opening **20**. In other embodiments, a spherical taper may be provided. The tapering portion **34** permits the generation of compression forces upon inserting a bone fastener through one of the openings **20**. The compression forces result in the sternum parts being drawn towards the fastening member **16** so as to increase the implant-bone-construct-stability.

(33) Further, each tapering portion **34** ends in a circumferential lip **36** of reduced material thickness around the respective hole **20**. The lip **36** constitutes a locking feature configured to lock a bone fastener to the fastening member **16** as generally illustrated in FIG. 6 (for a different cross-sectional configuration of the fastening member **16**).

(34) FIG. 6 illustrates a bone screw **40** with a shaft **42** and a head **44**. The shaft **42** of the bone screw **40** carries a bone thread **46** configured to engage bone. The head **44** of the bone screw **40** comprises a further thread **48** configured to engage the circumferential lip **36** at a selected angular orientation. As shown in FIG. 6, the selected angular orientation of the bone screw **40** may be within a range of 0° to 10° relative to the central axis of the opening **20**. Upon screwing the bone screw **40** into bone, the thread **48** at the head portion **44** of the bone screw **40** will engage the circumferential lip **36** to lock the bone screw **40** at the selected angular orientation to the fastening member **14**. As such, the thread **48** at the head **44** of the bone screw **40** constitutes a locking feature complementary to the circumferential lip **36** at the fastening member **16**.

(35) It will be appreciated that in other embodiments the circumferential lip **36** may be omitted or

replaced by another locking feature, such as a thread. It will also be appreciated that in other embodiments the thread **48** provided at the head **44** of the bone screw **40** could also be omitted. In such a case the head **44** of the bone screw **40** could have a smooth and spherically or conically tapering configuration.

(36) In the embodiment illustrated in FIG. 6, hole **20** tapers from a bone facing side of the fastening member **16** towards the lip **36** and also from the opposite side towards the circumferential lip **36**. One or both of those tapers could be omitted.

(37) Finally, FIGS. 7 and 8 illustrate the hook member **18** as well as the attachment of the hook member **18** to the elongated member **12**. As shown in FIG. 8, the hook member **18** comprises a sharpened tip **18A** at one end thereof as well as a structured region **18B** at the opposite end. The structured region **18B** facilitates injection molding of the elongated member **12** to the hook member **18**, as generally illustrated in FIG. 7.

(38) In the following, an exemplary method of using the implant **10** in connection with fixing two sternum parts will briefly be described. Similar methods may be applied in connection with other surgical indications for fixing bone parts in other regions of the human anatomy. In certain procedures, two sternum parts or portions may be separated by a bone cut. After these parts have been brought back together and preferably into their initial position with respect to one another, implant **10** is used for securing together these parts. U.S. Ser. No. 14/895,163 titled "Fixation Assembly with a Flexible Elongated Member for Fixing Parts of a Sternum," discloses in FIG. 10 thereof a fixation assembly attached to a sternum with bone parts having been brought back together and is hereby incorporated by reference herein in its entirety.

(39) In a first step, the elongated member **12** is wound around the two sternum parts that are to be fixed. Winding the elongated member **12** around the sternum parts is facilitated by the hook member **18** with the sharpened tip **18A**.

(40) In a next step the hook member **18** and then the elongated member **12** are threaded through the opening **30** of the engagement member **14**. In certain variants, the hook member **18** may be cut prior to that threading step, so that only the free end of the elongated member **12** (without the hook member **18**) is threaded through the corresponding opening **30**. Once the elongated member **12** enters the opening **30**, the engagement features **24** on the elongated member **12** are engaged by the complementary engagement feature **26** on the pawl **29** in a zip tie-like manner (i.e., such that the elongated member **12** can only be tightened, but cannot become loose).

(41) Then, the elongated member **12** is further drawn through the opening **30** so as to tighten the implant **10** around the two sternum parts. During that tightening process, the surgeon observes that the visual indicator **22** will substantially be placed on the joining line of the two sternum parts to enable a proper detachment. The tightening force is then increased further so as to compress the sternum parts together.

(42) If only a primary closure is needed, the above steps may be repeated one or multiple times for one or more further implants **10**. For primary closure, no further fixing of the implant **10** to the sternum parts using bone fasteners is needed.

(43) On the other hand, in case of a need for secondary closure, one or two fixation members are inserted in each bone part through the respective openings **20**. As has been explained above with reference to FIG. 6, each bone fasteners may be inserted at a selected angular orientation and may be locked to the implant **10** using the corresponding locking features **36**, **48**. The bone fasteners **40** may each have a self-drilling thread **46**. Alternatively, pilot holes may be drilled through the openings **20** prior to insertion of the bone fasteners **40**. In both cases, the thread **46** may be self-tapping.

(44) If a surgeon requires quick access to the thoracic region after the implant **10** has been implanted using one or more bone fasteners **40**, the surgeon uses a cutter to sever the fastening member **16** in the region of the visual indicator **22**. In a further step, the surgeon cuts the elongated member **12** so the sternum parts can be separated from each other. The bone fasteners **40** may

remain in place until after the thoracic region needs to be closed again via a primary or secondary closure process.

(45) The features described in the above description and illustrated in the accompanying drawings can be modified, replaced or combined to result in different embodiments. It will thus be apparent that the present disclosure may be varied in many ways. Such variations are not to be regarded as a departure from the scope of the present disclosure, and they are intended to be included within the scope of the following claims.

## Claims

1. An implant for bone fixation, comprising: an elongated member configured to be wound around bone parts that are to be fixed; an engagement member configured to engage a portion of the elongated member so as to secure the implant in a loop around the bone parts; and a fastening member comprising at least one opening for receiving a bone fastener, wherein the fastening member is coupled to at least one of the elongated member and the engagement member by injection molding.
2. The implant of claim 1, wherein the fastening member is coupled to the elongated member, the engagement member is coupled to the fastening member and the elongated member when the implant is secured in the loop around the bone parts.
3. The implant of claim 1, wherein the elongated member defines a first end and a second end, the fastening member is coupled to the second end of the elongated member, and the engagement member is spaced apart from the second end of the elongated member and is configured to engage a portion of the elongated member located between the first and second ends so as to secure the implant in the loop around the bone parts.
4. The implant of claim 1, wherein the elongated member comprises a plurality of engagement features provided along an extension of the elongated member, wherein the engagement member is configured to be brought into selective engagement with one of the engagement features.
5. The implant of claim 1, further comprising a hook member coupled to the elongated member at an end thereof opposite to the engagement member.
6. The implant of claim 5, wherein the hook member is coupled to the elongated member by injection molding.
7. The implant of claim 1, wherein the at least one opening of the fastening member comprises a locking feature configured to lock the bone fastener to the fastening member.
8. The implant of claim 1, wherein the fastening member is made from a metallic material and the elongate member is made from a polymeric material.
9. An implant system, comprising: an implant for bone fixation, comprising: an elongated member configured to be wound around bone parts that are to be fixed; an engagement member configured to engage a portion of the elongated member so as to secure the implant in a loop around the bone parts; and a fastening member comprising at least one opening for receiving a bone fastener, the fastening member being coupled to at least one of the elongated member and the engagement member by injection molding; and at least one bone fastener configured to be inserted through the at least one opening of the fastening member and into bone.
10. The implant system of claim 9, wherein the fastening member is coupled to the elongated member, the engagement member is coupled to the fastening member and the elongated member when the implant is secured in the loop around the bone parts.
11. The implant system of claim 9, wherein the elongated member defines a first end and a second end, the fastening member is coupled to the second end of the elongated member, and the engagement member is spaced apart from the second end of the elongated member and is configured to engage a portion of the elongated member located between the first and second ends so as to secure the implant in a loop around the bone parts.



12. The implant system of claim 9, wherein the at least one opening comprises a first locking feature configured to lock the bone fastener to the fastening member and wherein the bone fastener comprises a second locking feature configured to engage the first locking feature.
13. A method of manufacturing an implant, comprising: obtaining an elongated member configured to be wound around bone parts that are to be fixed; obtaining an engagement member configured to engage a portion of the elongated member so as to secure the implant in a loop around the bone parts; and coupling a fastening member having at least one opening for receiving a bone fastener to at least one of the elongated member and the engagement member via injection molding to form the implant.
14. The method of claim 13, wherein the fastening member is coupled to the elongated member, the engagement member is coupled to the fastening member and the elongated member when the implant is secured in the loop around the bone parts.
15. The method of claim 13, wherein the elongated member defines a first end and a second end, the fastening member is coupled to the second end of the elongated member, and the engagement member is spaced apart from the second end of the elongated member and is configured to engage a portion of the elongated member located between the first and second ends so as to secure the implant in the loop around the bone parts.
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